Hostile Intent Classifier

Project Proposal

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1. Problem Formulation:

Problem statement: General-purpose toxicity detectors are not always accurate when applied to in-game chat. They may rely heavily on profanity to identify toxicity, even though harassment does not necessarily contain profanity. At the same time, profanity between friends may be harmless banter rather than hostile speech. The meaning of a message can depend on its intent and conversational context. For example, a sarcastic comment from a stranger in a gaming lobby may be hostile, while a similar comment between friends may simply be playful banter.

1. Input and output:

   Input A: A message from an in-game chat, along with the surrounding conversation for context.

   Output A: A classification of whether the message contains hostile/toxic intent or non-hostile speech/banter.

2. What kind of task is this?

   This is a classification task. The model needs to classify in-game speech as either hostile or non-hostile while considering the presence of profanity and the conversational context of the message.

3. What would be considered a correct output?

A correct output would classify a message as hostile when it shows hostile intent toward another player, even if it does not contain profanity. It should also classify non-hostile banter correctly, even when profanity is present.

4. Research question:

To what extent do general-purpose toxicity classifiers rely on the presence of profanity rather than hostile intent when applied to in-game chat, and does in-domain fine-tuning with conversational context recover the cases they miss?

2. Why does someone care about this?

1. This problem is primarily faced by Online Gamers, who play matches with or against other gamers on the network.
2. The current system either:
    a. Flags the user if their speech contains profanity, regardless of hostility.

    or

    b. Allows the user to report the user, and the user is only flagged after a certain number of flags.
3. Gaming communication contains many nuances in how players interact, such as sarcasm, banter, game-specific language, and differences in how friends and strangers

communicate. Studying these interactions could help us understand how models can better recognize intent and context, with the potential for these methods to later be applied to detecting hostile speech in other forms of online communication.

3. Potential Research or Commercial Use Case

Use Cases:

1. When a gaming company wants to enhance their reporting systems for players during match conversations to keep a safe and playful environment (or PG-friendly) in all matches (statistically over 70% of gamers quit certain games due to toxicity)

2. Proactive detection of unreported harassment. Because the model scores every message rather than only reported ones, it can surface players whose behavior nobody has flagged.

3. Player-facing filtering rather than punishment. An optional setting for the user that mutes hostile messages for the individual player without any moderation action. Less punitive, and it respects that some communities want trash talk.

4. The social media world can take use of the model where comments left under users' public posts can possess certain thoughts and expressions and be read as sarcastic, cheerful or hurtful against the individual(s), and with the help of our model, it can remove the comment automatically. The tool can be completely optional to have on based on the user

Potential research

1. How much context is actually needed. Our model measures the marginal value of k prior turns on implicit toxicity specifically.

2. Toxicity classifiers are widely used in production. Measuring how much of their apparent performance comes from profanity matching is useful independent of the game setting.

4. Dataset

The datasets we acquired are:

- CONDA: a CONtextual Dual-Annotated dataset for in-game toxicity understanding and detection
- **GameTox**

1. One example is an individual Dota 2 chat message, such as “ez mid.” The input is the chat utterance (optionally with previous messages for context), and the target is its `intentClass`, such as I = Implicit Toxicity, E = Explicit Toxicity, or O = Non-toxic/Other.

2. Format: The CONDA dataset is provided as CSV files containing chat text, labels, conversation/match IDs, player information, timestamps, and token-level annotations.

3. Size: CONDA contains approximately 45,000 utterances from 12,000 conversations across 1,900 Dota 2 matches. The uploaded training split contains about 27,000 utterances. GameTox contains approximately 53k in game utterances

Citations:

@inproceedings{weld-etal-2021-conda,
  title = "{CONDA}: a {CON}textual Dual-Annotated dataset for in-game toxicity understanding and detection",
  author = "Weld, Henry and Huang, Guanghao and Lee, Jean and Zhang, Tongshu and Wang, Kunze and Guo, Xinghong and Long, Siqu and Poon, Josiah and Han, Caren",
  booktitle = "Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021",
  month = aug,
  year = "2021",
  address = "Online",
  publisher = "Association for Computational Linguistics",
  url = "https://aclanthology.org/2021.findings-acl.213",

    doi = "10.18653/v1/2021.findings-acl.213",
    pages = "2406--2416",
}

@inproceedings{naseem2025gametox,
  title={GameTox: A Comprehensive Dataset and Analysis for Enhanced Toxicity Detection in Online Gaming Communities},
  author={Naseem, Usman and Shiwakoti, Shuvam and Shah, Siddhant Bikram and Thapa, Surendrabikram and Zhang, Qi},
  booktitle={Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)},
  pages={440--447},
  year={2025}
}

5. Evaluation

1. What would an ideal system look like in action?

An ideal system would recognize the intent behind a player's message rather than simply looking for offensive words. For example, if friends jokingly use profanity with each other, the system should recognize it as non-hostile banter. At the same time, a message like *"Nobody wants you here, just uninstall"* should be recognized as hostile even though it contains no profanity.

2. What does good performance mean?

Good performance means the system generally agrees with how a human would interpret the conversation. It should identify messages that are genuinely hostile without unnecessarily flagging normal gaming banter. A bad system would frequently flag harmless profanity or fail to recognize hostility simply because the message does not contain obvious offensive words.

3. What are some tricky situations?

The system may struggle with sarcasm, inside jokes between friends, ambiguous messages, game-specific slang, and messages whose meaning depends heavily on earlier conversation. It may also struggle with intentionally disguised insults, unusual spelling, different languages or dialects, and communication styles from games that were not represented in the training data.

6. Logistical Plan

Before the preliminary results presentation, we plan to:

1. Prepare and clean the CONDA dataset.
2. Test existing toxicity classifiers on the dataset.
3. Analyze where the classifiers make mistakes, especially with profanity and implicit hostility.
4. Begin fine-tuning our own model and compare the results.

Work Split:
One member will focus more on testing existing classifiers, while the other will focus on fine-tuning and analyzing the results.

4. Long-term goal

The end state we are working toward is a moderation component that scores hostile intent rather than matching a wordlist, and that produces a per-player hostility score for a match rather than a verdict on isolated messages.
What our steps are:

1. Combine per-message predictions into a per-player score within a match, using player ID and match ID. This would match the decision a real moderation system makes. We will attempt an alpha version this term if time allows, but our focus remains message-level, so any player-level score is a collection rather than a validated output.

2. Cross-game generalization. Train on Dota 2 chat and test on another game's chat to see whether the learned notion of hostility is game-specific or general.

3. Relationship modeling. Move from same-team versus opposing-team toward genuine speaker-addressee modeling, which would need a dataset with addressee annotation that does not currently exist publicly.